

# Software Engineering - Lab 6:

## Microservices Technology Analysis

**Objective:** In this assignment you will explore the bookstore-microservices project to understand how multiple microservice patterns and supporting technologies work together. You will investigate authentication, inter-service communication styles, resiliency patterns, and observability tooling. Your deliverable is a single, well-structured report that documents your findings, visualizations, and answers to the questions below.

### Prerequisites

- **Node.js 18+** and **npm** installed locally.
- **Docker Desktop** running for containerized dependencies (databases, message brokers).
- Access to the bookstore-microservices repository and its `.env` templates.
- **A diagramming tool** of your choice (e.g., Draw.io, Excalidraw, Lucidchart, Mermaid, PowerPoint) for sequence diagrams, flowcharts, and graphs.
- **A spreadsheet or plotting tool** (e.g., Excel, Google Sheets, matplotlib) to create comparison charts.

### Submission

Submit **one PDF report** that includes:

- A short introduction summarizing your exploration approach.
- All written answers, tables, diagrams, charts, and screenshots requested in this assignment.
- Clear section headings matching the structure below.
- Captions for every visual (diagram, chart, screenshot) explaining what it shows.
- Appendix (optional) for supporting logs or code snippets referenced in the main text.

### Before Starting: Start infrastructure

From the repository root, run `docker compose up -d` (or the provided script) to launch databases, RabbitMQ, Kafka, Grafana/Prometheus, and any other shared services.

## Part 1: Backend API Inventory Table

Create a table that lists **10 publicly exposed backend API endpoints** in the system. For each endpoint, include:

- The route to the endpoint (e.g. GET /api/products)
- Service: which microservice owns the endpoint.
- Inter-Service: identify any downstream services called (REST, gRPC, messaging). If none, specify N/A.
- Data Store: primary database or data store touched during the request. If none, specify N/A.
- Messaging: queues/topics published to or consumed within the request flow. If none, specify N/A.
- Purpose: one-sentence summary of what the endpoint does.

## Part 2: Technologies Analysis

Answer each prompt concisely, adding diagrams or charts where requested.

### 1. Server-Sent Events (SSE)

- Identify which service implements SSE to communicate with the frontend. Describe its endpoint and the event payload structure you observe.

### 2. JWT Authentication

- Draw a **sequence diagram** with the actors:

User → API Gateway → User Service → Product Service → Product Database.

Show the interactions when a user authenticates, receives a JWT, and then **submits a product creation request** using that token.

### 3. Circuit Breaker

- Using the function names as implemented in order-service, create a **flowchart** illustrating how the circuit breaker transitions between states during a product lookup call.
- Explain why a circuit breaker is necessary in this context and what failure it guards against.

### 4. Compensating Transaction

- Identify the specific routes (include HTTP method and path) that implement a compensating transaction. Briefly describe the compensating action.

### 5. gRPC Performance Comparison

- In order-service/src/services/orderService.js, note the gRPC call that logs processing time (around line 60).

- Run the service normally (with gRPC enabled) and record the logged processing times for **five** order creation attempts.
- Comment out the gRPC client call and replace it with the corresponding REST fallback (as indicated in the code comments). Repeat the five-run timing experiment.
- Plot both datasets in a **bar graph** comparing gRPC vs. REST performance. Include the graph and a short interpretation of the results.

#### 6. **RabbitMQ**

- List all routing keys or topics produced to RabbitMQ across the services. State which service(s) publish and which consume each message

#### 7. **Kafka**

- Repeat the RabbitMQ analysis for Kafka topics: list producers and consumers.
- Compare RabbitMQ and Kafka in this project. Highlight how each is used differently and the benefits of this dual-messaging approach.

#### 8. **Grafana Observability**

- Provide a screenshot of the Grafana dashboard panel that visualizes per-service execution times. Describe any performance hotspots or anomalies you observe.

Ensure all sections are clearly labeled in your report and that every visual aid is legible. Submit your PDF through the course LMS by the deadline provided by your instructor.

## Grading Rubric

| Part                                    | Criteria                                                                                                                            |                                                                                                                         |        | Points |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------|--------|
| 1. Backend API Inventory Table          | For each of the 10 endpoints:<br><i>Partial credit (0.5) may be awarded within a component if information is partially correct.</i> |                                                                                                                         |        | 30     |
|                                         | Component                                                                                                                           | Criteria                                                                                                                | Points |        |
|                                         | 1. Route & Service Identification                                                                                                   | Correct HTTP method and path (e.g., <code>POST /api/orders</code> ) AND correct owning microservice                     | 1.0    |        |
|                                         | 2. System Interactions                                                                                                              | Correctly identifies downstream calls (REST/gRPC/Messaging or N/A) AND correctly identifies primary data store (or N/A) | 1.0    |        |
|                                         | 3. Messaging & Purpose                                                                                                              | Correct RabbitMQ/Kafka topics (or N/A) AND clear, accurate one-sentence purpose                                         | 1.0    |        |
| Subtotal                                |                                                                                                                                     |                                                                                                                         | 30     |        |
| 2.1 Server-Sent Events (SSE)            | Correct service identified                                                                                                          |                                                                                                                         |        | 2      |
|                                         | Correct endpoint identified                                                                                                         |                                                                                                                         |        | 2      |
|                                         | Event payload structure clearly described                                                                                           |                                                                                                                         |        | 2      |
| Subtotal                                |                                                                                                                                     |                                                                                                                         | 6      |        |
| 2.2 JWT Authentication Sequence Diagram | All required actors included<br>(User → API Gateway → User Service → Product Service → Product Database)                            |                                                                                                                         |        | 3      |
|                                         | Authentication flow correctly illustrated (login + JWT issuance)                                                                    |                                                                                                                         |        | 3      |
|                                         | Authenticated product creation flow correctly illustrated                                                                           |                                                                                                                         |        | 2      |
|                                         | Diagram clarity, correctness, and labeling                                                                                          |                                                                                                                         |        | 2      |
| Subtotal                                |                                                                                                                                     |                                                                                                                         | 10     |        |
| 2.3 Circuit Breaker                     | Flowchart reflects actual function names from order-service                                                                         |                                                                                                                         |        | 3      |
|                                         | Correct state transitions (Closed → Open → Half-Open)                                                                               |                                                                                                                         |        | 4      |

|                                                |                                                                                     |            |
|------------------------------------------------|-------------------------------------------------------------------------------------|------------|
|                                                | Clear explanation of why circuit breaker is necessary and failure it guards against | 3          |
| <b>Subtotal</b>                                |                                                                                     | <b>10</b>  |
| <b>2.4 Compensating Transaction</b>            | Correct HTTP method and route(s) identified                                         | 3          |
|                                                | Compensating action clearly and accurately described                                | 3          |
| <b>Subtotal</b>                                |                                                                                     | <b>6</b>   |
| <b>2.5 gRPC vs REST Performance Comparison</b> | Five timing samples recorded for gRPC                                               | 3          |
|                                                | Five timing samples recorded for REST fallback                                      | 3          |
|                                                | Bar graph comparing both datasets included and legible                              | 3          |
|                                                | Clear and accurate interpretation of performance results                            | 3          |
| <b>Subtotal</b>                                |                                                                                     | <b>12</b>  |
| <b>2.6 RabbitMQ Analysis</b>                   | All routing keys/topics correctly listed                                            | 3          |
|                                                | Producers correctly identified                                                      | 2          |
|                                                | Consumers correctly identified                                                      | 3          |
| <b>Subtotal</b>                                |                                                                                     | <b>8</b>   |
| <b>2.7 Kafka Analysis &amp; Comparison</b>     | All Kafka topics correctly listed                                                   | 3          |
|                                                | Producers correctly identified                                                      | 2          |
|                                                | Consumers correctly identified                                                      | 2          |
|                                                | Clear comparison between RabbitMQ and Kafka usage in this project                   | 3          |
| <b>Subtotal</b>                                |                                                                                     | <b>10</b>  |
| <b>2.8 Grafana Observability</b>               | Screenshot of correct Grafana dashboard panel included                              | 3          |
|                                                | Correct identification of per-service execution times                               | 2          |
|                                                | Meaningful analysis of hotspots or anomalies                                        | 3          |
| <b>Subtotal</b>                                |                                                                                     | <b>8</b>   |
| <b>TOTAL</b>                                   |                                                                                     | <b>100</b> |

Good luck!